

18A

CHEMICAL THERMODYNAMICS

UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 6

1. State the second law of thermodynamics.
2. Define entropy (S) and identify the sign of ΔS from chemical equations or changes of state.
3. State the third law of thermodynamics.
4. Calculate ΔS_{rxn} from molar entropies.
5. Compare and contrast ΔS_{univ} , ΔS_{sys} , and ΔS_{surr} .
6. Define Gibbs free energy (G) and describe how the signs of ΔH and ΔS affect the sign of ΔG .
7. Relate the sign of ΔG to the favorability/spontaneity of a reaction.
8. Determine the equilibrium temperature of a reaction.
9. Apply the relationship between ΔG° and the equilibrium constant (K).

Entropy and Spontaneity

18.1

Much of the work of chemists is to *make predictions* and answer questions about physical and chemical processes.

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|--|--|
| • What <i>changes</i> in a chemical reaction? | Chemical equations |
| • In what <i>proportions/quantities</i> ? | Stoichiometry |
| • How <i>fast/slow</i> does a reaction happen? | Kinetics and k |
| • To <i>what extent</i> does a reaction occur? | Equilibrium and K |
| • Is <i>heat</i> absorbed or released during the reaction? | Enthalpy: ΔH_{rxn} and q_{rxn} |

The focus of this chapter (thermodynamics) is:

- | | |
|---|----------------|
| • Does a reaction/process happen <u>spontaneously</u> ? | Thermodynamics |
|---|----------------|

Entropy and Spontaneity

18.1

Spontaneous processes: those that occur, under the specified conditions, *without the continuous supply of an external force/influence*.

- Spontaneity does *not* relate to time.
- Some thermodynamically spontaneous process occur so slowly they cannot be observed in our lifetime or even several lifetimes.
- Both endothermic ($\Delta H_{\text{rxn}} > 0$) and exothermic ($\Delta H_{\text{rxn}} < 0$) processes can be spontaneous.

Nonspontaneous processes will never occur on their own no matter how long you wait.

Entropy and Spontaneity

18.1

Entropy (S) is a measure of the degree of disorder (chaos or randomness) in a system.

- Entropy describes the *distribution* of energy within a system.
- When $\Delta S < 0$
- When $\Delta S > 0$

Second Law of Thermodynamics

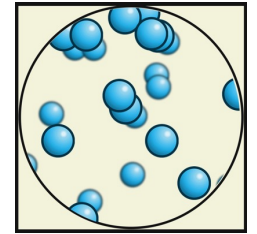
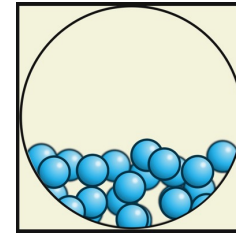
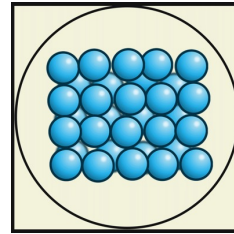
- A *spontaneous process* is one that results in an overall *increase* in the entropy (disorder) of the universe.

Entropy and Spontaneity

18.1

Factors that *affect the entropy* of a system:

1. Quantity or amount of substance
2. State of matter



3. Gases
4. Temperature

EXAMPLE PROBLEM

Learning Objective(s): 18.2

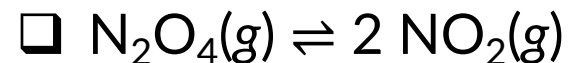
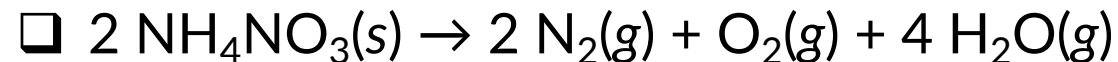
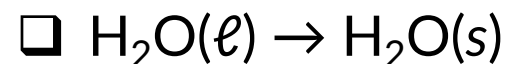
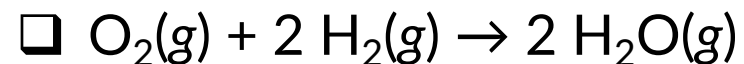
For each process, predict the sign of ΔS_{rxn} .

Reaction or Process	Sign of ΔS_{rxn}	
$\text{Ca}_3(\text{PO}_4)_2(s) \rightarrow 3 \text{Ca}^{3+}(aq) + 2 \text{PO}_4^{3-}(aq)$	$\Delta S_{\text{rxn}} < 0$	$\Delta S_{\text{rxn}} > 0$
$\text{I}_2(s) \rightarrow \text{I}_2(g)$	$\Delta S_{\text{rxn}} < 0$	$\Delta S_{\text{rxn}} > 0$
$2 \text{SO}_3(g) \rightleftharpoons 2 \text{SO}_2(g) + \text{O}_2(g)$	$\Delta S_{\text{rxn}} < 0$	$\Delta S_{\text{rxn}} > 0$
Freezing of water	$\Delta S_{\text{rxn}} < 0$	$\Delta S_{\text{rxn}} > 0$
Condensation	$\Delta S_{\text{rxn}} < 0$	$\Delta S_{\text{rxn}} > 0$

IN-CLASS QUESTION 1

Learning Objective(s): 18.2

For which process do you expect ΔS_{rxn} to be the *most positive*?



Entropy Changes

18.2

Third Law of Thermodynamics

- The absolute entropy of a pure, *perfectly ordered*, crystalline (solid) substance at absolute zero (0 Kelvin) is zero, $S = 0$.
- At absolute zero (0 K), there is no thermal energy (no motion).
- The third law provides a reference, a “zero point” ($S = 0$)!

Standard absolute molar entropies (S°)

- The absolute values of entropy (S) based on the “zero point” established by the third law of thermodynamics.
- Recall that the $^\circ$ symbol indicates standard conditions:
- This is unlike enthalpy where *only changes* (ΔH) could be measured.

Entropy Changes

18.2

Truncated table of standard molar entropies at 298 K (S°)

- These tables provided when needed, just like ΔH_f° .
- Greater values of S°

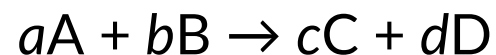
Substance (<i>state</i>)	$S^\circ \left(\frac{\text{J}}{\text{mol} \cdot \text{K}} \right)$
CO(g)	197.7
CO ₂ (g)	213.8
CaO(s)	38.1
CaCO ₃ (s)	91.7

Substance (<i>state</i>)	$S^\circ \left(\frac{\text{J}}{\text{mol} \cdot \text{K}} \right)$
H ₂ (g)	130.7
O ₂ (g)	205.2
I ₂ (g)	260.7

Substance (<i>state</i>)	$S^\circ \left(\frac{\text{J}}{\text{mol} \cdot \text{K}} \right)$
H ₂ O(<i>ℓ</i>)	70.0
H ₂ O(g)	188.8
Hg(<i>ℓ</i>)	75.9
Hg(g)	175.0
I ₂ (s)	116.2
I ₂ (g)	260.7

Entropy Changes

18.2



Calculating changes in enthalpy ($\Delta H_{\text{rxn}}^{\circ}$) from tabulated ΔH_f° values.

Calculating changes in entropy ($\Delta S_{\text{rxn}}^{\circ}$) from tabulated S° values.

IN-CLASS QUESTION 2

Learning Objective(s): 18.2, 18.4

Gaseous hydrogen and oxygen react to form water vapor.

a) Write a balanced chemical equation including phases.

b) For this reaction, it is predicted that: $\Delta S_{\text{rxn}}^{\circ} < 0$ $\Delta S_{\text{rxn}}^{\circ} > 0$

c) Calculate the standard change in entropy for this reaction ($\Delta S_{\text{rxn}}^{\circ}$) in $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$.

Substance (state)	$S^{\circ} \left(\frac{\text{J}}{\text{mol} \cdot \text{K}} \right)$
$\text{H}_2(\text{g})$	130.7
$\text{O}_2(\text{g})$	205.2
$\text{H}_2\text{O}(\ell)$	70.0
$\text{H}_2\text{O}(\text{g})$	188.8

BONUS QUESTION

Assessed on: Summer 2023, Final Exam

Select all processes for which you expect $\Delta S < 0$.

